higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

FITTING AND TURNING
NQF LEVEL 4

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This marking guideline consists of 6 pages.

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QUESTION 1: PUMPS AND COMPRESSORS

1.1
1.1.1 C
1.1.2 B
1.1.3 D
1.1.4 B
1.1.5 B
1.1.6 C

(6 × 1) (6)

1.2
- Ensure that the correct safety devices are installed.
- Do not operate the pump while valve inlets and outlets are closed.
- Do not use the pump for any other purpose than what it is designed for.
- Do not start the pump without priming it first or checking that it is primed. (4)

1.3
- Gear pump
- Helical gear screw pump
- Vane pump
- Flexible impeller pump (Any 2 × 1) (2)

1.4 Because failure to lock out driver power will result in serious injury. (1)

1.5 It cools the air thus reducing the volume before it enters the high pressure cylinder. (1 × 2) (2)

1.6
- Supply power to drive tool, such as pneumatic drills, jackhammers, rock drills and nail guns
- Supply power to high pressure cleaners, such as those used to blast debris off surfaces
- Providing the air to fill vehicle tyres
- Supply power to high pressure spray painters
- Controlling certain devices, such as the brakes on many vehicles (5)

[20]

QUESTION 2: HYDRAULICS AND PNEUMATICS

2.1
- Wash hands and any parts of the body that have been regularly in contact with hydraulic fluid.
- Wear protective gloves.
- Take care when working with cleaning and degreasing substances.
- Wear clean overalls on every shift.
- Put up appropriate warning signs when working with hydraulic equipment, and be mindful of your own safety as well as others. (5)

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2.2 2.2.1 Directional control valves ensure that fluid flows in the correct direction.

2.2.2 Safety valves are activated to protect the hydraulic system from being overloaded.

2.2.3 Variable flow valves maintain a constant flow rate and can be adjusted to different pressures.

(3 × 2) (6)

2.3
- Pneumatic system operates under air pressure, while hydraulic system operates under pressurised liquid.
- Pneumatic system works in an open–ended circuit, which means the air is released into the atmosphere, while hydraulic system works on a closed circuit where the fluid is directed back to the reservoir.
- Pneumatic system requires a larger cylinder than the hydraulic system to produce the same output.
- Pneumatic system uses a low-pressure fluid in comparison to the hydraulic system.
- Pneumatic system is much faster than the hydraulic system.
- Pneumatic system can operate under high temperatures while the hydraulic system cannot.
- Hydraulic systems are self-lubricating, while pneumatic system requires a lubricant for their moving parts.
- Pneumatic systems are more suitable for fire-risk areas, while hydraulic system can be very dangerous.
- Operating cost of a pneumatic system is much lower than operating costs for a hydraulic system. (Any 5 × 1) (5)

2.4
- Normally-closed valve is one that is closed under normal conditions and will open only when it is activated by air pressure.
- Normally-open valve is one that is open under normal conditions but will close, but not completely, when air pressure increases. (2)

2.5
- Kinks can cause pressure drops in the system.
- Kinks may also results in air leakage. (2)

[20]

QUESTION 3: SURFACE GRINDER

3.1
- Disc wheel or straight wheel
- Straight–cup grinding wheel
- Double–cup grinding wheel
- Flaring-cup grinding wheel
- Dish-grinding wheel (Any 4 × 1) (4)

3.2 3.2.1 Dressing the wheel – the purpose is to restore the cutting ability of the wheel.

3.2.2 Balancing the wheel – the purpose is in order to get a very fine and accurate finish and eliminate vibration of wheel.

(2 × 1) (2)

3.3
- Mount the diamond dresser onto the magnetic base and engage the chuck.
- Make sure that the coolant is on before the diamond touches the wheel.
- Get the diamond point to touch the wheel while it is revolving and move the table across the grinding wheel face.
- Move the diamond point clear of the grinding wheel, switch off the machine and remove the diamond point from the magnetic chuck. (4)

[10]

QUESTION 4: CENTRE LATHE MACHINE

4.1
- To protect lathe operator and other workers from injury.
- To ensure that you do not forget the chuck key in the chuck.
- To preserve/contain all cutting liquid in and around the lathe. (3)

4.2 Outside micrometer or vernier calliper (1)

4.3
- Fixed steady
- Travelling steady (2)

4.4 D = 200 mm
S = 250 mm/sec
N = ? r/min

$S = \pi \times D \times N$

$N = \frac{S}{\pi \times D}$

$= \frac{0,25 \times 60}{3,142 \times 0,2}$ ✓

= 23,87 r/sec✓
= 1432,209 r/min✓ (3)

4.5
- Whether or not the machined work piece matched the specification in the engineering drawings and job card.
- The cutters that were used for the machining job.
- The quality check and measuring instruments that were used. (3)

[12]

QUESTION 5: MILLING MACHINE

5.1
- Feed rate
- Material to be cut
- Cutting tool condition
- Cutting tool material
- The general depth of the cut should not exceed the diameter of the cut more than 1,5 times. (5)

5.2 Find the number of turns of the crank handle'

$$\text{Index} = \frac{40}{N}$$
$$= \frac{40}{58} \checkmark$$
$$= \frac{40 \div 2}{58 \div 2}$$
$$= \frac{20}{29} \checkmark$$

You will use plate 2 which contains a 29-hole circle. Movement will be 20 holes on a 29 hole-circle.✓ (3)

5.3
- End mill
- Face cutter
- Ball nose cutter
- Side cutter
- Slab cutter
- Radius cutter (Any 4 × 1) (4)

[12]

QUESTION 6: CNC MILLING AND CENTRE LATHE

6.1
- Remove all unused tools and store them in their assigned storage area.
- Remove all measuring equipment.
- Clean up any spill immediately.
- Store dirty, oily rags in a fireproof container.
- Sweep the floor around your work area. (Any 4 × 1) (4)

6.2 6.2.1 Cutting tool breaks rapidly or gets blunt very quickly due to overheating.

6.2.2 Time lost, low production rate

(2 × 1) (2)

6.3
- Import the geometry of the part to be manufactured.
- Identify the correct tool.
- Post the correct CNC programme to the correct machine. (3)

6.4
- Activate the desire programme that has to be altered.
- Ensure that the Program Protect key is switched to the off position.
- Scroll to the information that has to be changed.
- Type the correct value in the keypad.
- Press ALTER on the keypad to change the value. (5)

6.5 Solution:
#FL = 4
S = 68 m/min
Rpm = 720

$$MMP = \frac{m/min}{rpm \times \#FL}$$
$$= \frac{68}{720 \times 4}\checkmark$$
$$= \frac{68}{2880}\checkmark$$
= 0,0236 mm✓ (3)

6.6 6.6.1 Length = 110 mm✓
Width = 50 mm✓

6.6.2 S1200 indicate the cutting speed recommended for machine which is 1200 rpm.✓✓

6.6.3 N – Depicts (Indicates) the program line or block number of the written programme.✓
T – Code specifies the tool to be placed.✓
(3 × 2) (6)

6.7
- First–off or first–piece inspection/testing.
- During or in progress inspection/testing.
- Last-off, last piece or final inspection/testing. (3)

[26]

TOTAL: 100

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